

PROJECT I

Sedov Solution

When the shock wave sweeps-up enough space so that the interstellar medium mass many times exceeds the mass of supernova ejecta $4\pi R^3 \rho_o / 3 \gtrsim \mathcal{M}_o$, a supernova remnant enters the second phase of evolution. During this phase the shock wave decelerates according to the law $v_s = dR/dt = 2R/(5t)$, i.e.

$$R = \beta (E_o / \rho_o)^{1/5} t^{2/5}, \quad (1)$$

which with $\beta = 1.15$ is given in the analytical solution of Sedov (1959), for a point-like explosion with energy E_o in the uniform medium with density ρ_o . Sedov phase is approximately energy-conserving i.e. the expansion is nearly adiabatic. Independently of Sedov, this solution was found numerically by Taylor (1950), who used it to deduce the energy released in the Trinity atomic bomb test in Alamogordo, New Mexico. In 1950, by analyzing old photographs and applying his blast wave solution, Taylor calculated and published the explosion energy of the first atomic bomb. In the subsequent text, we shall explain how to obtain Sedov-Taylor solution numerically.

1. Euler Equations

Let us initially define the dimensionless quantities: radius $\xi = r/R$, density F , gas pressure G , cosmic ray pressure H , and velocity U :

$$\rho = \frac{\gamma + 1}{\gamma - 1} \rho_o F(\xi) \quad (2)$$

$$P_g = \frac{2}{\gamma + 1} \rho_o \dot{R}^2 G(\xi) \quad (3)$$

$$v = \frac{2}{\gamma + 1} \dot{R} U(\xi). \quad (4)$$

By using a procedur egiven in Arbutina (2017), from Euler equations

$$\frac{\partial \rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho v) = 0, \quad (5)$$

$$\rho \frac{Dv}{Dt} = \rho \frac{\partial v}{\partial t} + \rho v \frac{\partial v}{\partial r} = - \frac{\partial P_g}{\partial r}, \quad (6)$$

$$\frac{D(P_g \rho^{-\gamma})}{Dt} = \frac{\partial}{\partial t} (P_g \rho^{-\gamma}) + v \frac{\partial}{\partial r} (P_g \rho^{-\gamma}) = 0, \quad (7)$$

where $\frac{D}{Dt}$ is material derivative, we move to dimensionless equations

$$\left(\frac{U}{\xi} - \frac{\gamma+1}{2}\right)\xi\frac{F'}{F} + U' + \frac{2}{\xi}U = 0, \quad (8)$$

$$\xi\left[\left(\frac{2}{\gamma+1}\frac{U}{\xi} - 1\right)U' - \frac{3}{2}\frac{U}{\xi}\right] + \frac{\gamma-1}{\gamma+1}\frac{1}{F}G' = 0, \quad (9)$$

$$GF^{1-\gamma}\left(\frac{U}{\xi} - \frac{\gamma+1}{2}\right) = \frac{1-\gamma}{2\xi^3}, \quad (10)$$

In all the above expressions we have used a notation $f' = \frac{df}{d\xi}$, $f = \{F, G, U\}$.

If we use the adiabatic equation to find G' and insert it into momentum equation, with little transformation we obtain

$$\begin{aligned} \frac{dU}{d\xi} &= \frac{1}{\xi} \cdot \left\{ \frac{(\gamma-1)^2 F^{\gamma-2} \left(\frac{3}{2}(\gamma+1)\xi - 2\gamma U\right)}{(\gamma-1)^2 \gamma F^{\gamma-2} + 4\xi^2 \left(U - \frac{\gamma+1}{2}\xi\right)^3} \right. \\ &\quad \left. + \frac{3(\gamma+1)\xi^3 U \left(U - \frac{\gamma+1}{2}\xi\right)^2}{(\gamma-1)^2 \gamma F^{\gamma-2} + 4\xi^2 \left(U - \frac{\gamma+1}{2}\xi\right)^3} \right\}, \\ \frac{dF}{d\xi} &= -\frac{F}{U - \frac{\gamma+1}{2}\xi} \\ &\quad \times \left\{ \frac{\frac{3}{2}(\gamma+1)(\gamma-1)^2 F^{\gamma-2}}{(\gamma-1)^2 \gamma F^{\gamma-2} + 4\xi^2 \left(U - \frac{\gamma+1}{2}\xi\right)^3} \right. \\ &\quad \left. + \frac{\xi U (8U - (\gamma+1)\xi) \left(U - \frac{\gamma+1}{2}\xi\right)^2}{(\gamma-1)^2 \gamma F^{\gamma-2} + 4\xi^2 \left(U - \frac{\gamma+1}{2}\xi\right)^3} \right\}. \end{aligned} \quad (11)$$

These last two equations ought to be solved numerically. The results are shown in Fig. 1. As we approach the center $\xi \rightarrow 0$, the dimensionless velocity becomes $U \approx \frac{\gamma+1}{2\gamma}\xi$, the density $F \propto \xi^{3/(\gamma-1)}$ and the pressure $G \approx \text{const}$.

2. Energy integral

During the Sedov phase the energy is assumed to be conserved. If we want to find the total energy within the remnants E_o inside an expanding sphere defined with $\xi = 1$ (shock surface) it must be the sum of individual components – the kinetic and internal energy:

$$\begin{aligned} E_o &= T_o + U_o \\ &= \int_0^R \left(\frac{1}{2}\rho v^2 + \frac{1}{\gamma-1}P_g + \right) 4\pi r^2 dr \\ &= \frac{32\pi E_o}{25(\gamma+1)}\beta^5 \int_0^1 \left(\frac{1}{\gamma-1}FU^2 + \frac{1}{\gamma-1}G \right) \xi^2 d\xi. \end{aligned} \quad (12)$$

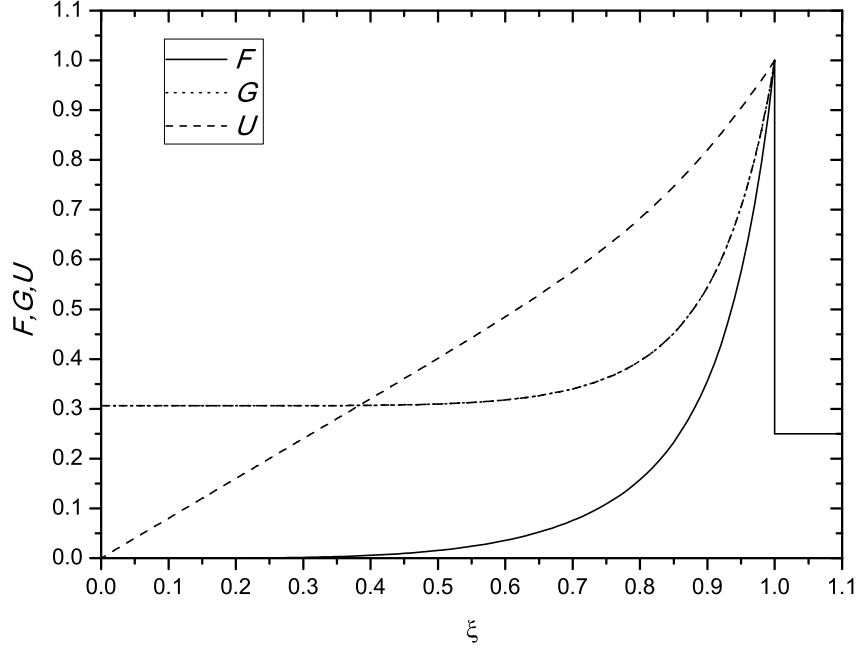


Figure 1: The Sedov solution for adiabatic index $\gamma = 5/3$.

The last expression allow us to find the parameter β :

$$\beta^{-5} = \frac{32\pi}{25(\gamma+1)} \int_0^1 \left(\frac{1}{\gamma-1} F U^2 + \frac{\beta}{\gamma-1} G + \frac{1-\beta}{\gamma_{CR}-1} H \right) \xi^2 d\xi, \quad (13)$$

as well as the fractions T_o/E_o , U_o/E_o .

3. The problem

For solving numerically dimensionless Euler equations we shall use the fourth order Runge-Kutta method. If we make a change of variables $x = 1-\xi$, $y = U$ and $z = F$, imamo :

$$y' = y'(x, y, z), \quad (14)$$

$$z' = z'(x, y, z). \quad (15)$$

The dimensionless pressure $g = G$ can be calculated from adiabatic equation for known y and z : $g = g(x, y, z)$. For known x_i , y_i , z_i at some point, with

some adapted small step $h = x_{i+1} - x_i$ one needs to calculate the quantities

$$\begin{aligned}
k_1 &= h y'(x_i, y_i, z_i), \\
l_1 &= h z'(x_i, y_i, z_i), \\
k_2 &= h y'(x_i + h/2, y_i + k_1/2, z_i + l_1/2), \\
l_2 &= h z'(x_i + h/2, y_i + k_1/2, z_i + l_1/2), \\
k_3 &= h y'(x_i + h/2, y_i + k_2/2, z_i + l_2/2), \\
l_3 &= h z'(x_i + h/2, y_i + k_2/2, z_i + l_2/2), \\
k_4 &= h y'(x_i + h, y_i + k_3, z_i + l_3), \\
l_4 &= h z'(x_i + h, y_i + k_3, z_i + l_3),
\end{aligned} \tag{16}$$

and the functions y_{i+1} i z_{i+1} at x_{i+1} we find as

$$y_{i+1} = y_i + \frac{k_1}{6} + \frac{k_2}{3} + \frac{k_3}{3} + \frac{k_4}{6}, \tag{17}$$

$$z_{i+1} = z_i + \frac{l_1}{6} + \frac{l_2}{3} + \frac{l_3}{3} + \frac{l_4}{6}. \tag{18}$$

At the shock ($x = 0$) the conditions are known: $y(0) = 1$, $z(0) = 1$, so we can use this as the starting point of integration, and stop when $x = 1$. With this approach try to reproduce Fig 1. Finally, find parameter β and fractions of kinetic and internal energy within the remnant.